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Arkansas Ecological Services Field Office

POPULATION STRUCTURE OF SELECTED FRESHWATER MUSSEL (BIVALVIA: UNIONOIDA) BEDS IN THE LITTLE RIVER, POND CREEK NATIONAL WILDLIFE REFUGE – PHASE I

Principal Investigator:

Chris L. Davidson
U.S. Fish and Wildlife Service
Arkansas Ecological Services Field Office
Conway, AR 72032
chris_davidson@fws.gov

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INTRODUCTION

The Little River in southeastern Oklahoma and southwestern Arkansas has a diverse mussel assemblage, including the federally protected Ouachita Rock Pocketbook (*Arcidens wheeleri*), Rabbitsfoot (*Quadrula cylindrica cylindrica*), Winged Mapleleaf (*Quadrula fragosa*), and Pink Mucket (*Lampsilis abrupta*). Several researchers over the past several decades have sampled mussels in much of the river (Ecosearch 1987; Harris and Gordon 1987; Galbraith *et al*. 2005, Vaughn 1994, 2012; Vaughn *et al*. 1995; Vaughn and Taylor 1999; Seagraves 2006; URS 2007; Arkansas Game and Fish Commission (AGFC) Mussel Database 2014; Galbraith and Vaughn 2011; Atkinson *et al*. 2012, 2014; Allen *et al*. 2013; Davidson *et al*. 2014). Mussel declines in the Little River largely have been attributed to impoundments (Vaughn and Taylor 1999; Galbraith and Vaughn 2011; Vaughn *et al*. 2015; Gates *et al*. 2015), drought (Atkinson *et al*. 2014; Vaughn *et al*. 2015), and degraded water quality from point source effluents (Ecosearch 1987). The factors limiting recruitment also include threats affecting their fish hosts (Haag and Warren 1997; Vaughn and Taylor 2000; Irmscher and Vaughn 2015).

Freshwater mussel species richness and community composition are influenced by numerous variables affecting habitat (e.g., land use, land cover, hydrology, etc.) at differing spatial scales (e.g., local to catchment) (Atkinson *et al*. 2012). Mussels do not occur continuously throughout the river, rather they occur in dense, multi-species aggregations called mussels beds (Strayer 2008). Mussel abundance and diversity are often greatest in medium to large rivers. Mussels are long-lived and relatively immobile. Therefore, they integrate environmental stressors occurring at temporal and spatial scales (local to catchment) (Atkinson *et al*. 2012).

Mussels have a larval stage that is an obligate ectoparasite on fish fins and gills (Barnhart *et al*. 2008; Haag 2012). Larvae metamorphose into juveniles that excyst from host fish and sink to the river bottom where they grow if habitat is suitable. Mussels may be fish host generalists (*e.g*., using many species) or specialists (*e.g*., using one species). Mussel beds represent local subpopulations linked to larger metapopulations through infrequent dispersal (Vaughn 1993, 2012; Newton *et al*. 2008). Mussel dispersal and gene flow is a function of host fish movement. Changes in flow regimes affect mussels and their fish hosts, dispersal potential, and population structure. Fish in the Little River move more upstream than downstream, allowing mussels to compensate for downstream displacement (Irmscher and Vaughn 2015).

Extensive systematic inventories of the Little River mussel fauna have been conducted from Pine Creek Lake to U.S. Highway 71 near Millwood Lake (Davidson *et al.* 2014; Vaughn 1994; Vaughn *et al.* 1995; Vaughn and Taylor 1999). Population structure of Little River mussel beds upstream of Millwood Lake in Arkansas, and more specifically Pond Creek National Wildlife Refuge (NWR), are lacking. Our survey presents baseline population structure of selected mussel beds in the Little River near or adjacent to the refuge. I am providing this information so future conservation efforts, including land management activities at Pond Creek NWR, can better protect the mussel fauna.

METHODS

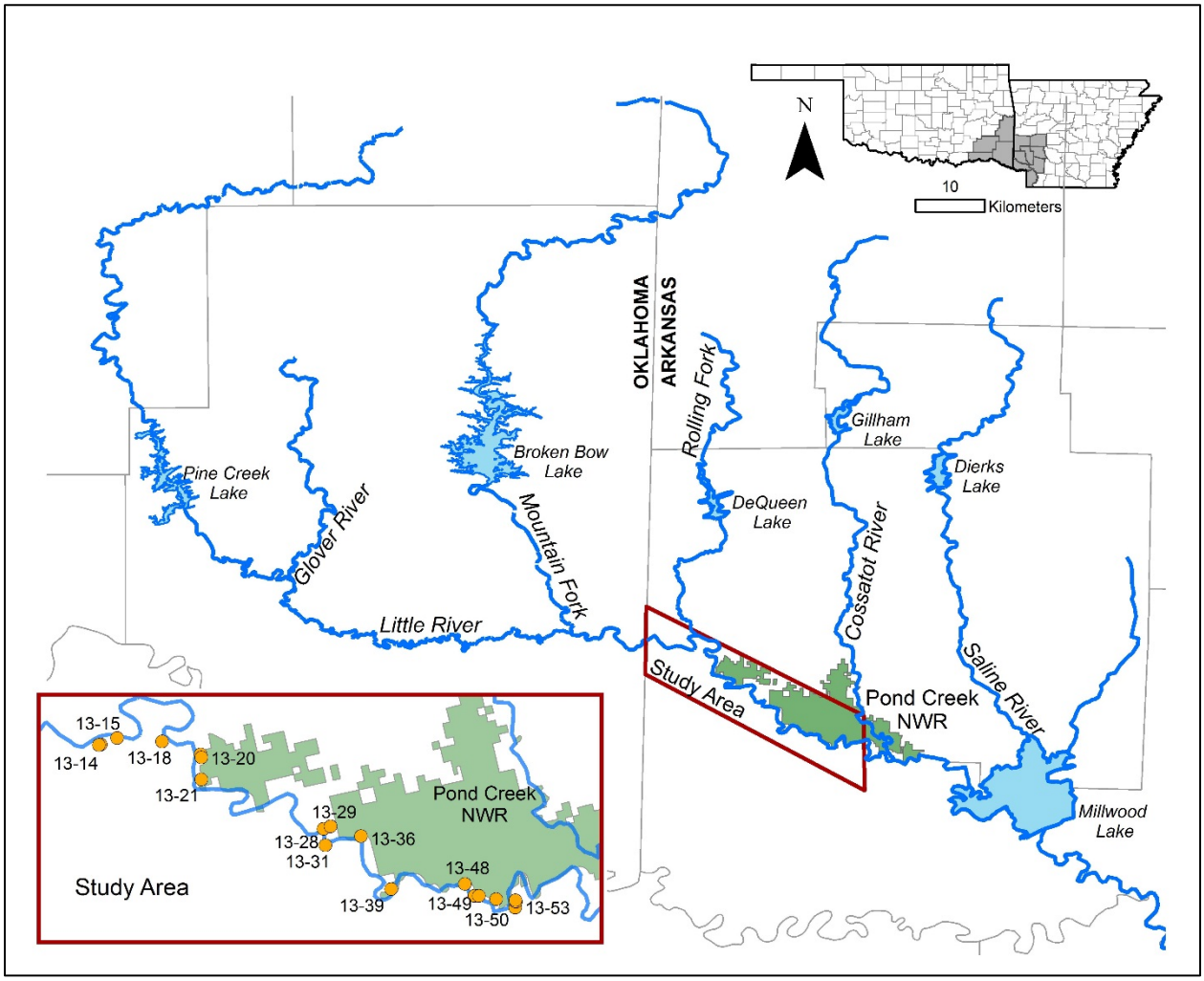
Study Area

The Little River originates near Ludlow in LeFlore County, Oklahoma, and flows in a southwesterly direction before entering Pine Creek Lake, one of two main stem impoundments. Upon exiting Pine

Creek Lake, the river flows in a southeasterly direction and enters Millwood Lake, approximately 56 km east of the Arkansas – Oklahoma state line, before converging with the Red River near Fulton in Hempstead County, Arkansas (Figure 1). The watershed drains an area of approximately 10,720 km^2 (Vaughn and Taylor 1999; Matthews *et al*. 2005). The headwaters and major tributaries are contained in the Ouachita Mountains and South Central Plains ecoregions, with the lower main stem wholly contained within the South Central Plains ecoregion (Woods *et al*. 2004). The area is characterized by a subhumid subtropical climate, mixed forests/woodlands, rugged mountains, broad valleys, and several large gravel bed rivers (OEAT 2003). Matthews *et al.* (2005) discuss physiography, climate, and land use in the watershed and geomorphology, hydrology, chemistry, biodiversity, and ecology of the Little River.

Three impoundments influence the Little River within our study area. The main stem of the river is impounded by the 1,644 km^2 Pine Creek Lake, which began operation in 1969, and the 118 km^2 Millwood Lake, which began operation in 1966. Pine Creek Lake, located approximately 93.5 km upstream of the Mountain Fork confluence is used for flood control, municipal water supply, and recreation. Outflow from Pine Creek Lake mimics inflow, but never drops below 1.4 m^3 s^{-1} to meet water quality criteria due to effluent from a paper mill (Galbraith and Vaughn 2011). The Mountain

FIGURE 1. Location of Little River study area and selected mussel beds sampled during 2016.



Fork, a major tributary of the Little River, is impounded by the 1,952 km^2 Broken Bow Reservoir, which was formed in 1968. Broken Bow Reservoir is used for flood control, hydropower, municipal water supply, and recreation. Cold water (hypolimnetic) outflow from Broken Bow Reservoir, located 32.7 km upstream of the Little River, contributes substantially to daily flow fluctuations in the Little River downstream of the Mountain Fork (Figure 2). The mussel community in the Little River downstream of the Mountain Fork confluence extending to the Arkansas – Oklahoma state line is affected by higher magnitude and colder flows than those of the natural flow regime (Galbraith and Vaughn 2011). Mussel species richness and abundance improves several km downstream from the state line (Davidson *et al*. 2014); however, outflow effects from Broken Bow Reservoir likely continue to exert influence on species presence/absence and life history requirements (e.g., host fish availability, reproduction, recruitment, etc.) of the Little River mussel fauna extending to Millwood Lake.

Pond Creek NWR includes a portion of the Little River within and along its southern boundary. The refuge boundary begins approximately 1.3 km downstream of the Arkansas Highway 41 Bridge and extends to Millwood Lake. The refuge boundary is interrupted by five "inholdings" or private tracts adjacent to the river (Davidson *et al.* 2014).

Survey Approach

In 2013, Davidson *et al*. (2014) delineated mussel beds in the Little River from the Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake. They also provide baseline semi-quantitative data for the mussel community in this reach of the Little River. Refuge management expressed a strong interest in determining population structure in selected Little River mussel beds delineated by Davidson *et al.* (2014). In our survey, 14 of 28 mussel beds delineated by Davidson *et al.* (2014) were selected for quantitative sampling based on presence of Pyramid Pigtoe (*Pleurobema rubrum*), a species petitioned for listing under the Endangered Species Act, and Ouachita Rock Pocketbook. One additional mussel bed quantitatively sampled in 2008 by U.S. Fish and Wildlife Service and Arkansas Game and Fish Commission staff is incorporated into this report (Table 1). Our survey represents the first quantitative sampling of mussel beds in the Arkansas portion of the Little River upstream of Millwood Lake. Mussel bed identification follows Davidson *et al*. (2014) (*e.g*., 13-XX).

Sampling was conducted from August 1 – October 6, 2016. Discharge (flow) was generally greater than the median discharge for the 48-year period of record during the survey time frame (Figure 2). Sample methodology for this monitoring effort uses methods of Christian and Harris (2005), which is a large black water mussel sampling modification of the stratified sampling design proposed by Strayer and Smith (2003). Christian and Harris (2005) found in large rivers where mussel densities average >10 mussels/m^2 that 15 1m^2 samples ensured 80% confidence in mussel bed species richness, mean species/m^2, and density estimates. They also found with mean mussel density >10/m^2 that 80% confidence was met using the Downing and Downing (1992) method. Strayer *et al.* (1997) and Vaughn *et al.* (1997) also found that sample size of 15 provides high confidence levels. Vaughn *et al.* (1997) showed that 10 0.25m^2 quadrats provided accurate estimates of the abundance of most species within mussel beds in the Little River basin. We sampled mussels generally in 1% of the bed area by excavating 1m^2 quadrats. Additional samples were collected in larger beds (> 10,000 m^2) to increase precision.

Mussel beds were accessed via boat. Mussel beds were searched from upstream to downstream to confirm and, if necessary, re-delineate the 2013 areal extent defined by Davidson *et al.* (2014). Each survey site was sampled using dive techniques and search strategies included hand grubbing and visual

searches. We first determined the extent of the mussel bed by diving over the area containing aggregated mussels and determining where mussel densities declined to < 10 mussels/m^2. We then measured mussel bed length using a Nikon rangefinder and width by the diver traversing from the inshore to mid-river limits in 1m increments. One m^2 quadrat sites were obtained using the Random# Generator application for iPhones and applied in an X, Y coordinate style. Mussels within the quadrat were collected by excavating the substrate and tactilely searching through the substrate to a depth of approximately 10 cm. Mussels were placed in a mesh bag and taken to the surface where they were identified, counted, and returned to the river. Pyramid Pigtoe, Ouachita Rock Pocketbook, and Rabbitsfoot were measured to the nearest 0.1 mm to provide baseline data on size frequencies for threatened, endangered, and petitioned species.

Nomenclature follows Turgeon *et al*. (1998), Inoue *et al*. (2013), Campbell and Lydeard (2012). The nomenclature for *Quadrula* is in flux at this time. We have chosen to reject changes within the *Quadrula* genus at this time until there is an empirical analysis of the *Quadrula* group as a whole.

We calculated species population estimates and assemblage total community numeric standing crop using the methods from Sampford (1962). The total number of mussels is:

$$X = \sum_{o}^{i} y_i \; x \; g_i$$

Figure 2. Little River discharge (flow) near Horatio, Arkansas, from August 1 – October 6, 2016 (USGS provisional data) https://nwis.waterdata.usgs.gov/ar/nwis/uv/?cb_00060=on&format=gif_default&site_no=07340000&period=&begin_date=2016-08-01&end_date=2016-10-06.

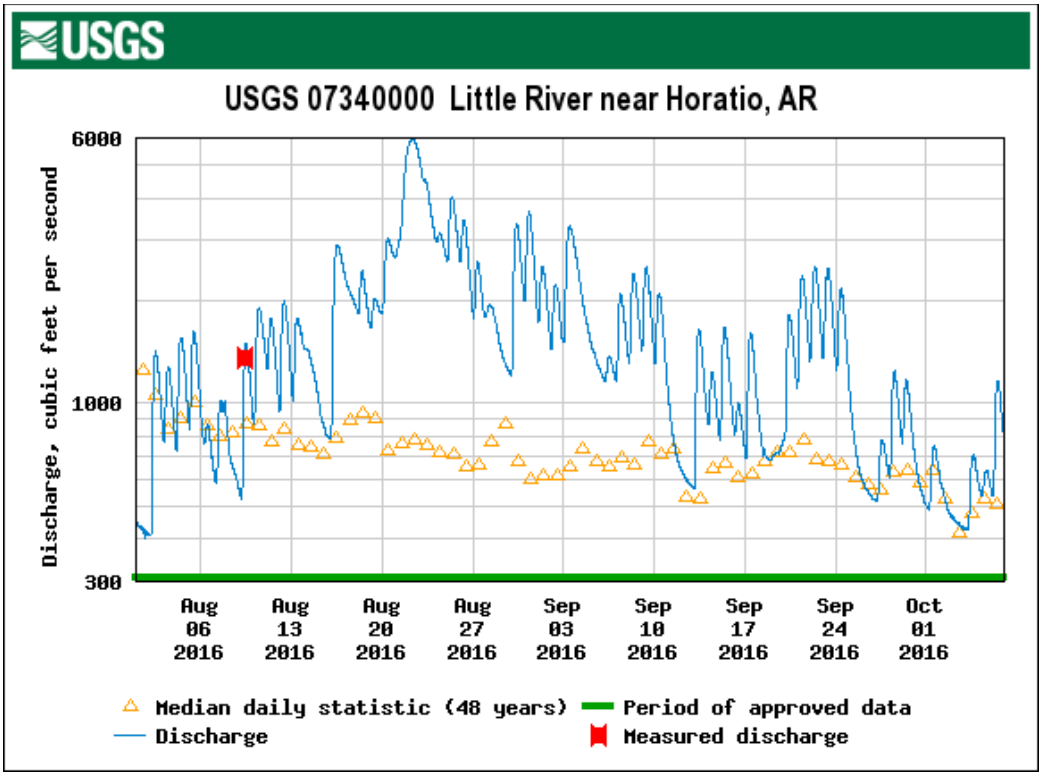


Table 1. Little River, Arkansas – Oklahoma state line to Millwood Lake, mussel bed locations (Davidson *et al.* 2014). Yellow highlighted mussel beds sampled during current survey. Green highlighted mussel bed sampled during 2008. * (1), (2), (3) denotes strata

Mussel Bed ID	Start		End	
	Latitude	Longitude	Latitude	Longitude
13-03	33.93918	-94.47459	33.93951	-94.47315
13-07	33.94117	-94.46233	NR	NR
13-08	33.93490	-94.45818	33.93463	-94.45775
13-12	33.92895	-94.44717	33.92958	-94.44569
13-14(1)*	33.93041	-94.43978	33.93038	-94.44097
13-14(2)	33.93038	-94.44097	33.93065	-94.43872
13-15	33.93137	-94.42947	33.93070	-94.42861
13-20(1)	33.91321	-94.38457	33.91162	-94.38466
13-20(2)	33.91162	-94.38466	33.90845	-94.38661
13-21	33.90123	-94.38792	33.90055	-94.38890
13-23(1)	33.88988	-94.36750	33.88938	-94.36626
13-23(2)	33.88938	-94.36626	33.88871	-94.36444
13-23(3)	33.88871	-94.36444	33.88806	-94.36263
13-27	NR	NR	33.86074	-94.32953
13-28	33.86252	-94.32603	33.86344	-94.32494
13-29	33.86281	-94.32163	33.86177	-94.31990
13-31	33.85450	-94.32745	33.85352	-94.32664
13-17	33.92646	-94.41564	33.92648	-94.41500
13-18	33.92412	-94.40460	33.92383	-94.40450
13-19	33.92297	-94.40281	33.92299	-94.40197
13-36	33.85428	-94.30586	33.85178	-94.30172
13-39	33.82551	-94.29681	33.82498	-94.29843
13-40	33.82174	-94.30362	33.82122	-94.30270
13-43	33.82965	-94.28702	33.82907	-94.27991
13-44	33.82216	-94.27566	33.82197	-94.27303
13-45	33.81724	-94.26824	33.81780	-94.26297
13-48	33.81855	-94.25420	33.81295	-94.25107
13-49(1)	33.81179	-94.25060	33.81114	-94.24812
13-49(2)	33.81114	-94.24812	33.80975	-94.24432
13-50	33.80742	-94.23877	33.80292	-94.23873
13-53(1)	33.80107	-94.22925	33.80373	-94.22806
13-53(2)	33.80373	-94.22806	33.80453	-94.22812
13-53(3)	33.80453	-94.22812	33.80752	-94.22938
13-54	33.80941	-94.22845	33.81107	-94.22447
13-57	33.80227	-94.21610	33.80297	-94.21349

Where X is total number of mussels in a bed, *i* is the number of strata, y_i is the sample total (total number of organisms encountered in the n_i sampling units) and g_i is the raising factor ($g_i = 1/f_i$,) where f_i is the fraction sampled and is defined by n_i/N_i, with n_i being the number of sample units counted in the *i*th stratum, and N_i the total potential number of sampling units in the *i*th stratum (Huebner *et al.* 1990).

The 95% confidence interval around the total number of mussels in a mussel bed was calculated using:

$$X \pm \left(t \; x \sqrt{\sum_0^i N_i^2 \; x \; S^2 \; y_i \; x \; \frac{1-f_i}{n_i}} \right)$$

Where S^2y_i is the sample variance computed from raw counts in the n_i sampling units in the *i*th stratum and t is the Student's *t* for effective degrees of freedom (Huebner *et al.* 1990).

We used the following formula to estimate the relative error of our sample size in determining mean species density and to determine the number of quadrat samples required to estimate mean species density with 95 percent confidence limits (Southwood 1978):

$$n = (s + (Ex))^2$$

where:

n = sample size (number of quadrats to sample)
s = standard deviation of mussel density
E = predetermined standard error as a decimal
x = mean mussel density

It was not an objective of this survey to ascertain whether or not outflows from Broken Bow Reservoir or Pine Creek Lake are affecting mussel assemblages in our study area. However, we believe that our data present baseline information that can be used to further validate and delineate main stem and tributary impoundment effects to the Little River mussel fauna described in Vaughn and Taylor (1999). More importantly, we believe these data can be used as a baseline comparison to evaluate the success of future mussel conservation efforts in the Little River.

RESULTS

A total of 14 mussel beds were quantitatively sampled in 2016 (Table 2 and Appendices A1 – A14) and one mussel bed in 2008 (Appendix B). Mean mussel bed area was 6,327m^2 with a range of 720 – 15,525m^2 (Table 2). Since the Davidson *et al*. (2014) delineation of mussel beds in 2013, areal extent decreased at Site 13-14, 13-15 and 13-31 by 4,175, 1,530 and 360m^2, respectively (Table 2).

Twenty-six (26) species were recorded during our sampling effort (Table 3). Species richness increased with increasing mussel bed size (Figure 3). Six species (*Elliptio dilatata, Lampsilis teres, Lasmigona complanata, Quadrula metanevra, Q. fragosa,* and *Strophitus undulatus*) documented by Davidson *et al.* (2014) were not collected during this study. All species documented during our study were collected by Davidson *et al.* (2014), including the federally endangered Ouachita Rock Pocketbook and federally threatened Rabbitsfoot (Table 3). Mean species richness for all beds was 18 with a range of 9 – 25 (Appendices A1 – A14 and B).

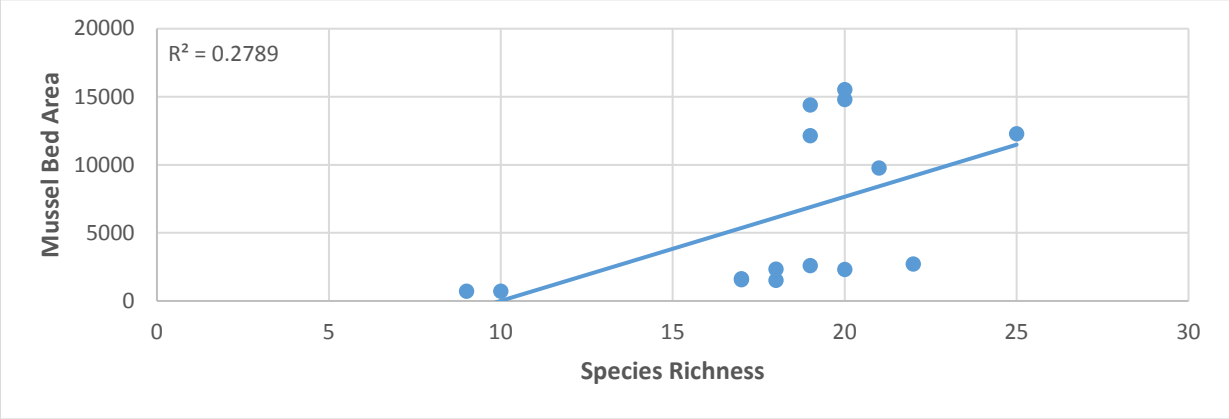
Species composition and dominance varied little across sampled mussel beds. Four species comprised 88 percent of the total mussel community. The remaining 22 species each comprised ≤1.3% of the total mussel community. Pimpleback (*Quadrula pustulosa*) and Mucket (*Actinonaias ligamentina*) were the

two dominant species comprising 50 and 15% of the total, respectively. Threehorn Wartyback (*Obliquaria reflexa*) and Deertoe (*Truncilla truncata*) comprised the remainder of the 88% (13 and 11 percent, respectively). Pimpleback was the dominant species in all mussel beds.

Table 2. Area of selected mussel beds delineated by Davidson *et al*. (2014) in 2013 and re-delineated in our study (2016) in the Little River, Arkansas.

Mussel Bed	Davidson *et al.* (2014) Mussel Bed Area (m^2)	Current Study (2016) Mussel Bed Area (m^2)
13-14	6,475	2,300
13-15	2,250	720
13-18	1,625	1,625
13-20	12,280	12,280
13-21	1,500	1,500
13-27	15,525	NA
13-28	720	720
13-29	2,700	2,700
13-31	1,920	1,560
13-36	14,400	14,400
13-39	2,600	2,600
13-48	14,800	14,800
13-49	9,760	9,760
13-50	2,340	2,340
13-53	12,130	12,130

Figure 3. Mussel species—area relationship for select mussel beds in the Little River, Arkansas.



A total of six Ouachita Rock Pocketbook, 47 Rabbitsfoot, and 29 Pyramid Pigtoe individuals were collected in our survey (Appendices A1 – A14 and B). Ouachita Rock Pocketbook was collected in 4 of 15 mussel beds (13-20, 13-21, 13-27, and 13-53). Our sampling methodology underestimated the relative abundance of Rabbitsfoot. Rabbitsfoot relative abundance was generally greater in areas with lower water velocity near the periphery of the wetted channel and upstream margin of shoals, areas that also tended to have < 10 mussels/m^2. For example, 59 Rabbitsfoot were collected in 0.2 person hour near the right-hand margin and upstream extent of the shoal at Site 13-21 compared to two individuals collected in quadrat sampling at this site.

Approximate age of Ouachita Rock Pocketbook individuals collected in 2016 (n = 5) was six – nine years based on external growth rings (annuli). Pyramid Pigtoe and Rabbitsfoot were collected in 11 of 15 and 12 of 15 mussel beds, respectively. Mean length of rabbitsfoot was 89.8 mm (SD = 9.8) with a range of 58.4 – 107.6 mm. Mean height of Ouachita Rock Pocketbook was 54.4 mm (SD = 7.4) with a range of 44.0 – 63.3 mm. Mean height of Pyramid Pigtoe was 57.8 (SD = 6.8) with a range of 46.3 – 70.4 mm.

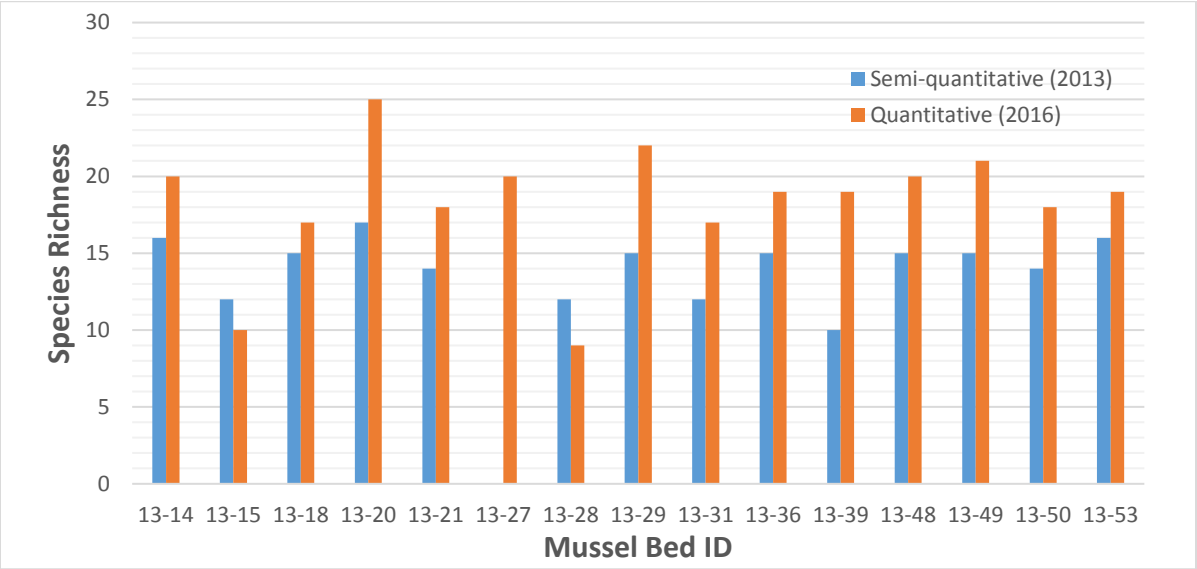
Table 3. Mussel species (n = 26) collected in quadrat samples from select mussel beds in the Little River between the Arkansas – Oklahoma state line and U.S. Highway 71, 2016.

Species	**Common Name**
Actinonaias ligamentina	Mucket
Amblema plicata	Threeridge
Arcidens wheeleri	Ouachita Rock Pocketbook
Ellipsaria lineolata	Butterfly
Fusconaia flava	Wabash Pigtoe
Lampsilis cardium	Plain pocketbook
Lampsilis sp. B cf. *hydiana*	Louisiana Fatmucket
Lasmigona costata	Fluted-shell
Leptodea fragilis	Fragile Papershell
Megalonaias nervosa	Washboard
Obliquaria reflexa	Threehorn Wartyback
Obovaria arkansasensis	Southern Hickorynut
Plectomerus dombeyanus	Bankclimber
Pleurobema sintoxia	Round Pigtoe
Pleurobema rubrum	Pyramid Pigtoe
Pleurobema riddelli	Louisiana Pigtoe
Potamilus purpuratus	Bleufer
Ptychobranchus occidentalis	Ouachita Kidneyshell
Quadrula apiculata	Southern Mapleleaf
Quadrula cylindrica cylindrica	Rabbitsfoot
Quadrula nobilis	Gulf Mapleleaf
Quadrula pustulosa	Pimpleback
Reginaia ebena	Ebonyshell
Tritogonia verrucosa	Pistolgrip
Truncilla donaciformis	Fawnsfoot
Truncilla truncata	Deertoe

Quantitative sampling in our survey generally detected greater species richness in mussel beds compared to semi-quantitative sampling conducted by Davidson *et al*. (2014), the exception being Sites 13-15 and 13-28 (Figure 4). The areal extent of Site 13-15 during our survey was 68% smaller than 2013, which may account for lower species richness. Quadrat sampling in our survey failed to detect one to three rare species (relative abundance < 1%) documented by Davidson *et al*. (2014) in 8 of 14 mussel beds, excluding Site 13-15 and 13-18 where the number of species not detected was five and seven, respectively. Quadrat sampling failed to detect our target species (Pyramid Pigtoe and Ouachita Rock Pocketbook) at Site 13-18 and 13-50. In general, the species we failed to detect in quadrat samples occur in (1) near bank or shoal habitats that may have been under sampled due to randomly selected sample location and (2) areas not meeting our criteria for mussel bed areal limits. These data indicate

that both semi-quantitative and quadrat sampling should be used to more precisely assess the distribution and richness of species.

Figure 4. Species richness in selected Little River mussel beds in Arkansas, 2013 and 2016. Site 13-27 was sampled once in 2008.



Appendix C figures show a regression of cumulative number of species versus the log of cumulative individuals for each mussel bed to predict effort required to obtain an additional species. Based on these data, two to eight additional samples were required to detect an additional species at 12 of 14 sites. Site 13-29 and 13-48 required 20 and 68 additional samples, respectively, to detect an additional species.

A total of 10,658 individuals were collected from 360 1m^2 samples. Mean (SD), minimum and maximum mussel density and relative abundance for each species is reported in Appendices A1 – A14 and B. Community numeric standing crop ranged from 7,097 ± 1,712 (Site 13-15) – 592,482 ± 90,971 (Site 13-20). Confidence levels for estimating mean mussel density (individuals/m^2) using 1m^2 quadrats ranged from 83 – 94% with a mean confidence level of 90% (SD = 3.0). The number of quadrats required to achieve a 95% confidence level ranged from 35 – 164 with a mean of 91 (SD = 39.6). I believe these data indicate sufficient statistical power to detect future changes in species relative abundance and community numeric standing crop.

DISCUSSION

Mussel assemblages typically consist of a few dominant, common species and many more less abundant or rare species (Vaughn 1997). The majority of mussel species (approximately 80%) make up less than 10% of mussel beds and a substantial number (approximately 35%) comprise less than one percent. It is rare for a single species to compose greater than 50% of a mussel bed (Haag 2012). I observed similar results in the 15 mussel beds sampled during this survey. Species abundance in mussel beds in the Kiamichi River (Little River basin) exhibit similar patterns to what I observed in this survey; the four most abundant species compose >70% of the community (Spooner and Vaughn 2009). Consistent with other mid- to large-sized streams in the Mississippian region (Haag 2012), stream size generalists dominated mussel beds in this survey.

Atkinson *et al.* (2012) found a longitudinal shift in dominant mussel species in the Little River upstream of Pine Creek Lake, with community composition strongly correlated to distance from the headwaters (*i.e*., stream size). Haag's (2012) synthesis of similar studies also describes a predictable increase in species richness along stream size gradients in the Mississippian region. Atkinson *et al*. (2012) found species composition was structured by landscape factors after accounting for stream size. Our survey sites were close spatially (39.5 km between the most upstream and most downstream mussel beds), in stream size (all sites except 13-14 and 13-15 were located between the Rolling Fork and Cossatot River confluences), and land use was similar (Figure 1). As such, a longitudinal shift in species composition and dominance was not evident in our survey area.

Vaughn and Taylor (1999) found a linear effect gradient in the Little River downstream of Pine Creek Lake and the Mountain Fork confluence (Broken Bow Reservoir), with mussel richness and abundance increasing with distance from the dam. They did not survey downstream of the Arkansas – Oklahoma state line and as such did not observe a mussel "recovery" in the 15 km reach downstream of the Mountain Fork confluence. The mussel "recovery" did not occur until approximately 20 km downstream of Pine Creek Lake (Vaughn and Taylor 1999). The likely mechanism, supported by evidence of reduced gametogenesis in *Quadrula*, underlying the decline in mussel density and occurrence downstream of the Mountain Fork is reduced reproduction due to high discharge and abnormally cold water during summer spawning periods (Galbraith and Vaughn 2009, 2011). Davidson *et al*. (2014) and this survey document increases in mussel species richness and abundance downstream of the Arkansas – Oklahoma state line. There appears to be a similar 20 km distance necessary downstream of both Pine Creek and Broken Bow dam inflows to the Little River for mussel "recovery".

An increase in species richness with increasing habitat area (species—area relationship) is a fundamental pattern in ecology. Larger areas support more species presumably because they support a greater diversity of habitats that support larger populations with lower probabilities of local extirpation (Rosenzweig 1995, Gotelli and Graves 1996 *in* Haag 2012). Local assemblages for many animals are structured spatially by small-scale variables in habitat. Several mussel species are microhabitat specialists, occurring under large, nonembedded slab rocks, in aquatic vegetation such as water willow (*Justicia americana*), or predominantly in slow current adjacent to the bank or swift currents. Rabbitsfoot and Pistolgrip (*Tritogonia verrucosa*) were the only species encountered during our survey that exhibit known, specialized microhabitat requirements. Our results also indicate that species such as Ouachita Rock Pocketbook, mischaracterized as an oxbow and slack water specialist in the literature (U.S. Fish and Wildlife Service 2004), can inhabit main channel mussel beds, albeit typically restricted to near bank areas on the inside of river meanders. We found mussel community composition in our study area to be similar among sites with site specific variation in species richness, at least in part, attributable to mussel bed size and the presumably greater diversity of habitats provided by larger mussel beds.

The relative immobility, long lifespan, and reproductive characteristics of mussels make them vulnerable to habitat disturbance (Strayer 2008). Substrate stability, particularly shear stress at high flow, has been shown to be strongly correlated and predictive of mussel occurrence in all stream sizes (Haag 2012; Peck 2005; Morales *et al*. 2006). These stable flow refuges may persist for decades allowing large numbers of individuals to accumulate (*i.e*., mussel bed). Habitat stability helps explain the lack of consistency in microhabitat associations across stream types in the literature (Haag 2012). Therefore, disturbance by high flows, drought, and a variety of other factors play a large role in determining mussel occurrence. I found two sites (13-14 and 13-15) where available habitat (bed area) substantially decreased since 2013, potentially indicating these sites are becoming unstable and may not persist

much longer. Site 13-15 has an eroding stream bank with denuded riparian cover on river left that likely is contributing to instability of the site.

Complex hydraulic conditions during extreme events (floods, droughts) have a stronger influence on the distribution of adult mussels than average hydraulic conditions (Gangloff and Feminella 2007, Zigler *et al*. 2008, Allen and Vaughn 2010). Allen *et al*. (2013) found climate variables were not important in predicting low flow days in the Little River due to dam releases. Decreases in water temperature downstream of hypolimnetic release dams have been shown to reduce and even eliminate mussel populations for long distances (*in* Gates *et al*. 2015). Release of cold water during summer when water temperature should be warm suppresses mussel metabolic rates during a time of year when growth should be high (McMahon and Bogan 2001) and inhibits reproduction (Layzer *et al*. 1993). Cold water releases also may inhibit or eliminate reproduction of warm water fishes (Layzer *et al.* 1993; Yeager 1993). Abnormally cold discharge during summer may act as a permanent colonization barrier to non-generalist mussels (Vaughn and Taylor 1999).

Pine Creek Lake releases generally mimic inflows during the entire year, with slight deviations in February and March. Broken Bow releases vary considerably from inflow with reservoir releases substantially higher than inflow during summer months when natural stream flow is low and water temperature warmer. Mussel populations in the Little River (Oklahoma) downstream of the Mountain Fork confluence experience significantly colder than normal summer water temperatures, higher flow and potentially limited food (phytoplankton) availability compared to sites downstream of Pine Creek Lake (Galbraith and Vaughn 2009, 2011). Disruptions to mussel reproductive cues downstream of dams have resulted in reproductive failure (Layzer *et al*. 1993; Heinricher and Layzer 1999). Galbraith and Vaughn (2011) found mussel populations at a Little River site downstream of the Mountain Fork exhibited greater stress, higher rates of parasitism, lower body condition, a higher frequency of hermaphroditism, and evidence that lower water temperatures may be disrupting "typical" season patterns in gametogenesis. While mussel species richness and abundance increases 20 km downstream of the Mountain Fork confluence, it remains unknown whether mussel condition and reproductive traits continue to be disrupted in our survey area.

Constraints to mussel recovery include time lags associated with their long life span and reduced reproduction and dispersal related to negative density dependence and river fragmentation. Species densities depend strongly on past, as well as current ecological conditions, and it can take decades for species to recover from environmental change (Strayer 2008). Long-term persistence depends on migration of new individuals with different genetic material between local populations. For mussels, this migration must occur mainly via movement of fish hosts, and successful colonization of new patch should be a function of both the ability to infect the host and the host's distribution, abundance and movement. As local populations decline or are eliminated and dispersal distance increases, demographic and genetic constraints will diminish the ability of currently abundant species to respond to natural disturbance and much less to anthropogenic change (Vaughn 1993).

MANAGEMENT CONSIDERATIONS AND MONITORING RECOMMENDATIONS

Protection or improvements in the status of mussels in the Little River requires proper management of the watershed, reservoirs, and cooperative efforts of stakeholders. The following management and monitoring recommendations are provided to assist with future refuge management.

Management Considerations

Flow Regimes

1. Flow regimes downstream of Broken Bow Reservoir decrease mussel species richness and abundance for at least 20 km downstream of the Mountain Fork confluence. Because mussels are dependent on host fish, any effects of hydrological alteration on host fish also affects mussel populations. Persistence of the Little River mussel fauna downstream of the Mountain Fork may require decreasing hypolimnetic releases at Broken Bow Reservoir from May to September to encompass temperature dependent spawning needs of both short- and long-term brooders (Galbraith and Vaughn 2009). Short-term brooders spawn, brood, and release glochidia over a 2 – 6 week time period in spring/summer, while long-term brooders spawn in late summer/fall, brood eggs over winter and release glochidia in the spring/summer. Therefore, aligning Broken Bow releases to prioritize seasonal environmental flows (discharge and temperature) should reduce mussel and host fish mortality and optimize their recruitment.

2. Model simulations indicate dam operations probably affect juvenile settling by altering complex hydraulic parameters during periods when recently metamorphosed juveniles release from fish (Daraio *et al.* 2010). Studies of juvenile settling are providing crucial information about how flow alterations can directly and indirectly optimize mussel bed formation and persistence (Daraio *et al.* 2010). Juvenile settling work is also being used to inform models that simulate mussel population dynamics incorporating biotic interactions and abiotic conditions (Morales *et al.* 2006).

3. Evaluate whether land use in the Rolling Fork watershed and dam releases at DeQueen Lake are significant factors in mussel distribution and abundance in the Little River downstream of the Rolling Fork confluence.

4. Identify other important factors affecting mussel dispersal, spatial distribution patterns, and genetic connectivity in the Little River.

5. Global climate change and water diversion for human use will likely continue to affect mussel distribution and abundance in the Little River basin. Increasing riparian forests will help minimize water temperature warming and reduce evaporation rates. I recognize this management consideration may seem contrary to Number 1, but riparian forests also act to stabilize riverine habitats which is vital to mussel colonization and persistence.

6. Implement best management practices to reduce sediment runoff from construction sites, county road maintenance, and other land use activities (*e.g*., forestry) to further improve water quality and habitat availability in the Little River.

Execution of these management considerations and strategies should improve and/or sustain the quality and quantity of vegetative cover in riparian areas, decrease siltation, provide environmental flows that promote mussel recruitment and survival and subsequently improve habitat quality.

Monitoring Recommendations

1. I recommend monitoring the status of mussel distribution and abundance in the Little River at least once per decade. Sampling techniques should include both semi-quantitative (e.g., timed transect searches, etc.) and quantitative (e.g., $1m^2$ quadrat) methods.

2. More frequent monitoring may be necessary to evaluate changes in mussel distribution, abundance, and recruitment to reduce uncertainties about how mussels respond to changing flow conditions or other management considerations.

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Appendix A1 – A14. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A1			
Location: 13-14		Date: 06 September 2016	
Estimated Mussel Bed Area (m^2):		2,300	
1 m^2 Quadrats Sampled:		23	
Min – Max Density (#/m^2):		11 – 50	
Mean Density #/m^2 (SD):		24.5 (11.4)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	59	10.5	5,900 ± 2,632
Amblema plicata	2	0.4	200 ± 238
Ellipsaria lineolata	1	0.2	100 ± 168
Fusconaia flava	4	0.7	400 ± 412
Lampsilis cardium	9	1.6	900 ± 657
Lasmigona costata	2	0.4	200 ± 238
Leptodea fragilis	7	1.2	700 ± 468
Megalonaias nervosa	7	1.2	700 ± 394
Obliquaria reflexa	20	3.6	2,000 ± 853
Pleurobema riddellii	7	1.2	700 ± 532
Pleurobema rubrum	1	0.2	100 ± 168
Pleurobema sintoxia	13	2.3	1,300 ± 793
Potamilus purpuratus	5	0.9	500 ± 437
Pytchobranchus occidentalis	1	0.2	100 ± 168
Quadrula cylindrica cylindrica	11	2.0	1,100 ± 714
Quadrula nobilis	2	0.4	200 ± 238
Quadrula pustulosa	314	55.8	31,400 ± 4,247
Tritogonia verrucosa	10	1.8	1,000 ± 662
Truncilla donaciformis	1	0.2	100 ± 168
Truncilla truncata	87	15.5	8,700 ± 2,912
Totals	563	100.3	56,300 ± 9,627

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A2			
Location: 13-15		Date: 06 September 2016	
Estimated Mussel Bed Area (m^2):		720	
1 m^2 Quadrats Sampled:		7	
Min – Max Density (#/m^2):		5 – 16	
Mean Density #/m^2 (SD):		9.9 (3.3)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	8	11.6	823 ± 420
Amblema plicata	2	2.9	206 ± 251
Ellipsaria lineolata	1	1.4	103 ± 192
Obliquaria reflexa	6	8.7	617 ± 462
Pleurobema sintoxia	2	2.9	206 ± 387
Quadrula cylindrica cylindrica	1	1.4	103 ± 192
Quadrula nobilis	1	1.4	103 ± 192
Quadrula pustulosa	44	63.8	4,526 ± 1,315
Tritogonia verrucosa	1	1.4	103 ± 192
Truncilla truncata	3	4.3	309 ± 279
Totals	69	99.8	7,097 ± 1,712

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A3			
Location: 13-18		Date: 03 August 2016	
Estimated Mussel Bed Area (m^2):		1,625	
1 m^2 Quadrats Sampled:		16	
Min – Max Density (#/m^2):		9 – 50	
Mean Density #/m^2 (SD):		24.6 (13.8)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	54	13.7	5,484 ± 2,219
Amblema plicata	11	2.8	1,117 ± 629
Ellipsaria lineolata	10	2.5	1,016 ± 582
Fusconaia flava	5	1.3	508 ± 346
Lampsilis cardium	2	0.5	203 ± 250
Obliquaria reflexa	55	14.0	5,586 ± 2,357
Obovaria arkansasensis	2	0.5	203 ± 250
Plectomerus dombeyanus	5	1.3	508 ± 433
Pleurobema riddellii	3	0.8	305 ± 289
Pleurobema rubrum	1	0.3	102 ± 177
Pleurobema sintoxia	12	3.0	1,219 ± 617
Potamilus purpuratus	6	1.5	609 ± 445
Quadrula apiculata	1	0.3	102 ± 177
Quadrula nobilis	3	0.8	305 ± 395
Quadrula pustulosa	179	45.4	18,180 ± 4,317
Tritogonia verrucosa	5	1.3	508 ± 433
Truncilla truncata	40	10.2	4,036 ± 1,688
Totals	394	100.2	40,016 ± 9,989

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A4			
Location: 13-20		Date: 02 August 2016	
Estimated Mussel Bed Area (m^2):		1.) 2,280 2.) 10,000 Total: 12,280	
1 m^2 Quadrats Sampled:		40	
Min – Max Density (#/m^2):		13 - 101	
Mean Density #/m^2 (SD):		1.) 34.7 (15.3) 2.) 51.3 (26.6) Total: 48.4 (25.6)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	223	11.5	68,166 ± 13,169
Amblema plicata	13	0.7	3,985 ± 2,086
Arkansia wheeleri	3	0.2	932 ± 964
Ellipsaria lineolata	59	3.0	18,060 ± 5,480
Fusconaia flava	43	2.2	13,166 ± 5,195
Lampsilis cardium	7	0.4	2,121 ± 1,795
Lasmigona costata	2	0.1	606 ± 798
Leptodea fragilis	3	0.1	932 ± 1,260
Megalonaias nervosa	16	0.8	4,894 ± 2,928
Obliquaria reflexa	200	10.3	61,105 ± 11,792
Obovaria arkansasensis	4	0.2	1,257 ± 1,078
Plectomerus dombeyanus	9	0.5	2,750 ± 1,740
Pleurobema ridellii	9	0.5	2,727 ± 1,890
Pleurobema rubrum	8	0.4	2,424 ± 1,244
Pleurobema sintoxia	55	2.8	16,848 ± 5,124
Potamilus purpuratus	27	1.4	8,273 ± 3,339
Ptychobranchus occidentalis	5	0.3	1,515 ± 1,198
Quadrula apiculata	3	0.2	909 ± 961
Quadrula cylindrica cylindrica	8	0.4	2,424 ± 2,021
Quadrula nobilis	5	0.3	1,538 ± 1,214
Quadrula pustulosa	931	48.1	284,685 ± 55,999
Reginaia ebena	2	0.1	606 ± 798
Tritogonia verrucosa	14	0.7	4,288 ± 2,110
Truncilla donaciformis	1	0.1	303 ± 573
Truncilla truncata	287	14.8	87,968 ± 17,313
Totals	1,937	100.1	592,482 ± 90,971

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A5			
Location: 13-21		Date: 01 August 2016	
Estimated Mussel Bed Area (m^2):		1,500	
1 m^2 Quadrats Sampled:		15	
Min – Max Density (#/m^2):		9 – 37	
Mean Density #/m^2 (SD):		24.0 (7.1)	
Species	Number Collected	Percent of Total	Population Estimate
Actinonaias ligamentina	47	13.1	4,700 ± 2,082
Amblema plicata	12	3.3	1,200 ± 690
Arkansia wheeleri	1	0.3	100 ± 183
Ellipsaria lineolata	8	2.2	800 ± 617
Fusconaia flava	4	1.1	400 ± 309
Lampsilis cardium	3	0.8	300 ± 309
Lasmigona costata	1	0.3	100 ± 183
Leptodea fragilis	1	0.3	100 ± 183
Obliquaria reflexa	63	17.5	6,300 ± 1,363
Obovaria jacksoniana	2	0.6	200 ± 218
Pleurobema rubrum	1	0.3	100 ± 183
Pleurobema sintoxia	2	0.6	200 ± 218
Potamilus purpuratus	2	0.6	200 ± 218
Ptychobranchus occidentalis	1	0.3	100 ± 183
Quadrula cylindrica cylindrica	2	0.6	200 ± 378
Quadrula pustulosa	162	45.0	16,200 ± 3,024
Tritogonia verrucosa	2	0.6	200 ± 218
Truncilla truncata	46	12.8	4,600 ± 1,746
Totals	360	100.3	36,000 ± 4,876

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A6			
Location: 13-28		Date: 07 September 2016	
Estimated Mussel Bed Area (m^2):		720	
1 m2 Quadrats Sampled:		7	
Min – Max Density (#/m^2):		14 – 39	
Mean Density #/m^2 (SD):		23.9 (10.3)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	8	4.8	823 ± 548
Amblema plicata	3	1.8	309 ± 404
Ellipsaria lineolata	2	1.2	206 ± 251
Obliquaria reflexa	26	15.6	2674 ± 643
Plectomerus dombeyanus	1	0.6	103 ± 192
Qudrula cylindrica cylindrica	3	1.8	309 ± 1276
Quadrula pustulosa	110	65.9	11,314 ± 4,735
Tritogonia verrucosa	1	0.6	103 ± 192
Truncilla truncata	13	7.8	1,337 ± 909
Totals	167	100.1	17,177 ± 5,268

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A7			
Location: 13-29		Date: 08 September 2016	
Estimated Mussel Bed Area (m^2):		2,700	
1 m^2 Quadrats Sampled:		25	
Min – Max Density (#/m^2):		9 – 52	
Mean Density #/m^2 (SD):		29.6 (9.6)	
Species	2016 Number Collected	2016 Percent of Total	2016 Population Estimate
Actinonaias ligamentina	66	8.9	7,128 ± 1,690
Amblema plicata	18	2.4	1,944 ± 732
Arkansia wheeleri	1 FD		NA
Ellipsaria lineolata	24	3.2	2,592 ± 1,157
Fusconaia flava	5	0.7	540 ± 518
Lampsilis cardium	1	0.1	108 ± 189
Lampsilis Sp. B	1	0.1	108 ± 189
Lasmigona costata	1	0.1	108 ± 189
Megalonaias nervosa	5	0.7	540 ± 423
Obliquaria reflexa	146	19.7	15,768 ± 3,367
Obovaria arkansasensis	3	0.4	324 ± 299
Plectomerus dombeyanus	2	0.3	216 ± 267
Pleurobema ridellii	1	0.1	108 ± 189
Pleurobema rubrum	2	0.3	216 ± 267
Pleurobema sintoxia	5	0.7	540 ± 423
Potamilus purpuratus	7	0.9	756 ± 423
Qudrula cylindrica cylindrica	1	0.1	108 ± 189
Quadrula nobilis	1	0.1	108 ± 189
Quadrula pustulosa	375	50.6	40,500 ± 5,669
Reginaia ebena	1	0.1	108 ± 189
Tritogonia verrucosa	14	1.9	1,512 ± 668
Truncilla truncata	62	8.4	6,696 ± 1,817
Totals	741	99.8	80,028 ± 9,028

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A8			
Location: 13-31		Date: 09 September 2016	
Estimated Mussel Bed Area (m^2):		1560	
1 m^2 Quadrats Sampled:		17	
Min – Max Density (#/m^2):		9 – 83	
Mean Density #/m^2 (SD):		38.6 (22.0)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	82	12.5	7,107 ± 2,115
Amblema plicata	6	0.9	520 ± 445
Ellipsaria lineolata	7	1.1	607 ± 395
Fusconaia flava	5	0.8	433 ± 373
Lampsilis cardium	3	0.5	260 ± 252
Megalonaias nervosa	4	0.6	347 ± 356
Obliquaria reflexa	99	15.1	8,580 ± 2,553
Obovaria arkansasensis	3	0.5	260 ± 252
Plectomerus dombeyanus	8	1.2	693 ± 507
Pleurobema rubrum	1	0.2	87 ± 159
Pleurobema sintoxia	2	0.3	173 ± 205
Potamilus purpuratus	7	1.1	607 ± 455
Ptychobranchus occidentalis	1	0.2	87 ± 159
Qudrula cylindrica cylindrica	1	0.2	87 ± 159
Quadrula pustulosa	367	55.9	31,807 ± 9,525
Tritogonia verrucosa	8	1.2	693 ± 331
Truncilla truncata	53	8.1	4,593 ± 1,453
Totals	657	98.6	56,940 ± 15,061

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A9			
Location: 13-36		Date: 15 September 2016	
Estimated Mussel Bed Area (m^2):		14,400	
1 m^2 Quadrats Sampled:		30	
Min – Max Density (#/m^2):		0 – 49	
Mean Density #/m^2 (SD):		22.5 (14.4)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	112	16.6	53,760 ± 19,059
Amblema plicata	1	0.1	480 ± 796
Ellipsaria lineolata	19	2.8	9,120 ± 3,845
Fusconaia flava	7	1.0	3,360 ± 2,055
Lampsilis cardium	2	0.3	960 ± 1126
Lasmigona costata	1	0.1	480 ± 796
Megalonaias nervosa	1	0.1	480 ± 796
Obliquaria reflexa	74	10.9	35,520 ± 12,072
Obovaria arkansasensis	1	0.1	480 ± 796
Plectomerus dombeyanus	3	0.4	1,440 ± 1,379
Pleurobema rubrum	1	0.1	480 ± 796
Pleurobema sintoxia	3	0.4	1,440 ± 1,379
Potamilus purpuratus	11	1.6	5,280 ± 2,907
Ptychobranchus occidentalis	1	0.1	480 ± 796
Qudrula cylindrica cylindrica	4	0.6	1,920 ± 2,055
Quadrula nobilis	2	0.3	960 ± 1126
Quadrula pustulosa	364	53.8	174,720 ± 38,532
Tritogonia verrucosa	13	1.9	6,240 ± 3,845
Truncilla truncata	56	8.3	22,080 ± 9,305
Totals	676	99.5	324,480 ± 66,135

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A10			
Location: 13-39		Date: 16 September 2016	
Estimated Mussel Bed Area (m^2):		2,600	
1 m^2 Quadrats Sampled:		25	
Min – Max Density (#/m^2):		4 – 54	
Mean Density #/m^2 (SD):		24.8 (13.4)	
Species	Number Collected	Percent of Total	Population Estimate
Actinonaias ligamentina	87	14.1	9,048 ± 2,034
Amblema plicata	10	1.6	1,040 ± 643
Ellipsaria lineolata	20	3.3	2,080 ± 997
Fusconaia flava	4	0.6	416 ± 407
Lampsilis cardium	8	1.3	832 ± 575
Leptodea fragilis	1	0.2	104 ± 182
Obliquaria reflexa	118	19.0	12,272 ± 3,570
Obovaria arkansasensis	5	0.8	520 ± 498
Plectomerus dombeyanus	2	0.3	208 ± 257
Pleurobema ridellii	2	0.3	208 ± 257
Pleurobema rubrum	1	0.2	104 ± 182
Potamilus purpuratus	11	1.8	1,144 ± 575
Ptychobranchus occidentalis	1	0.2	104 ± 182
Qudrula cylindrica cylindrica	1	0.2	104 ± 182
Quadrula nobilis	1	0.2	104 ± 182
Quadrula pustulosa	257	41.5	26,728 ± 6,880
Quadrula verrucosa	4	0.6	416 ± 288
Truncilla donaciformis	1	0.2	104 ± 182
Truncilla truncata	85	13.7	8,840 ± 2,054
Totals	619	100.1	64,376 ± 12,202

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A11			
Location: 13-48		Date: 26 September 2016	
Estimated Mussel Bed Area (m^2):		14,800	
1 m^2 Quadrats Sampled:		30	
Min – Max Density (#/m^2):		10 – 58	
Mean Density #/m^2 (SD):		30.6 (12.0)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	185	20.1	9,1267 ± 16,225
Amblema plicata	13	1.4	6,413 ± 2,987
Ellipsaria lineolata	39	4.2	19,240 ± 7,317
Fusconaia flava	10	1.1	4,933 ± 2,987
Lampsilis cardium	1	0.1	493 ± 818
Lasmigona costata	1	0.1	493 ± 818
Megalonaias nervosa	1	0.1	493 ± 818
Obliquaria reflexa	110	12.0	54,267 ± 10,667
Obovaria arkansasensis	2	0.2	493 ± 818
Plectomerus dombeyanus	3	0.3	1,480 ± 2,112
Pleurobema ridellii	16	1.7	7,893 ± 3,340
Pleurobema rubrum	3	0.3	1,480 ± 1,494
Pleurobema sintoxia	6	0.7	2,960 ± 2,112
Potamilus purpuratus	1	0.1	493 ± 818
Qudrula cylindrica cylindrica	4	0.4	1,973 ± 2,059
Quadrula nobilis	3	0.3	1,480 ± 1,494
Quadrula pustulosa	452	49.2	222,987 ± 35,378
Reginaia ebena	1	0.1	493 ± 818
Tritogonia verrucosa	2	0.2	987 ± 1157
Truncilla truncata	66	7.2	32,560 ± 8,449
Totals	919	99.9	453,375 ± 56,464

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A12			
Location: 13-49		28/29 September 2016	
Estimated Mussel Bed Area (m^2):		1.) 6,160 2.) 3,600 Total: 9,760	
1 m^2 Quadrats Sampled:		25	
Min – Max Density (#/m^2):		8 – 59	
Mean Density #/m^2 (SD):		1.) 38.6 (11.9) 2.) 29.4 (12.9) Total: 35.3 (12.8)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	102	11.6	39,821 ± 10,945
Amblema plicata	11	1.2	4,294 ± 2,167
Ellipsaria lineolata	35	4.0	13,664 ± 5,735
Fusconaia flava	8	0.9	3,123 ± 1,877
Lampsilis cardium	2	0.2	781 ± 1,084
Leptodea fragilis	1	0.1	390 ± 685
Megalonaias nervosa	1	0.1	390 ± 685
Obliquaria reflexa	96	10.9	37,478 ± 8,464
Obovaria arkansasensis	1	0.1	390 ± 685
Plectomerus dombeyanus	8	0.9	3,123 ± 2,655
Pleurobema ridellii	8	0.9	3,123 ± 2,167
Pleurobema rubrum	2	0.2	781 ± 970
Pleurobema sintoxia	8	0.9	3,123 ± 1,877
Potamilus purpuratus	10	1.1	3,904 ± 1,877
Ptychobranchus occidentalis	2	0.2	781 ± 969
Quadrula apiculata	1	0.1	390 ± 685
Qudrula cylindrica cylindrica	4	0.5	1,562 ± 1,084
Quadrula nobilis	2	0.2	781 ± 969
Quadrula pustulosa	506	57.3	197,542 ± 25,919
Tritogonia verrucosa	5	0.6	1,952 ± 1877
Truncilla truncata	70	7.9	27,328 ± 6,502
Totals	883	100.0	344,723 ± 43,914

Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

Appendix A13			
Location: 13-50		Date: 5 October 2016	
Estimated Mussel Bed Area (m^2):		2,340	
1 m^2 Quadrats Sampled:		23	
Min – Max Density (#/m^2):		7 – 49	
Mean Density #/m^2 (SD):		23.7 (12.7)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	105	19.3	10,683 ± 2,201
Amblema plicata	5	0.9	509 ± 445
Ellipsaria lineolata	11	2.0	1,119 ± 623
Fusconaia ebena	1	0.2	102 ± 171
Fusconaia flava	7	1.3	712 ± 541
Lampsilis cardium	1	0.2	102 ± 171
Lasmigona costata	1	0.2	102 ± 171
Megalonaias nervosa	1	0.2	102 ± 171
Obliquaria reflexa	88	16.2	8,953 ± 3,004
Obovaria arkansasensis	3	0.6	305 ± 296
Plectomerus dombeyanus	1	0.2	102 ± 171
Pleurobema sintoxia	1	0.2	102 ± 171
Potamilus purpuratus	3	0.6	305 ± 296
Ptychobranchus occidentalis	3	0.6	305 ± 296
Quadrula pustulosa	276	50.7	28,080 ± 6,903
Tritogonia verrucosa	4	0.7	407 ± 331
Truncilla donaciformis	1	0.2	102 ± 171
Truncilla truncata	32	5.9	3,256 ± 990
Totals	544	100.2	55,346 ± 10,842

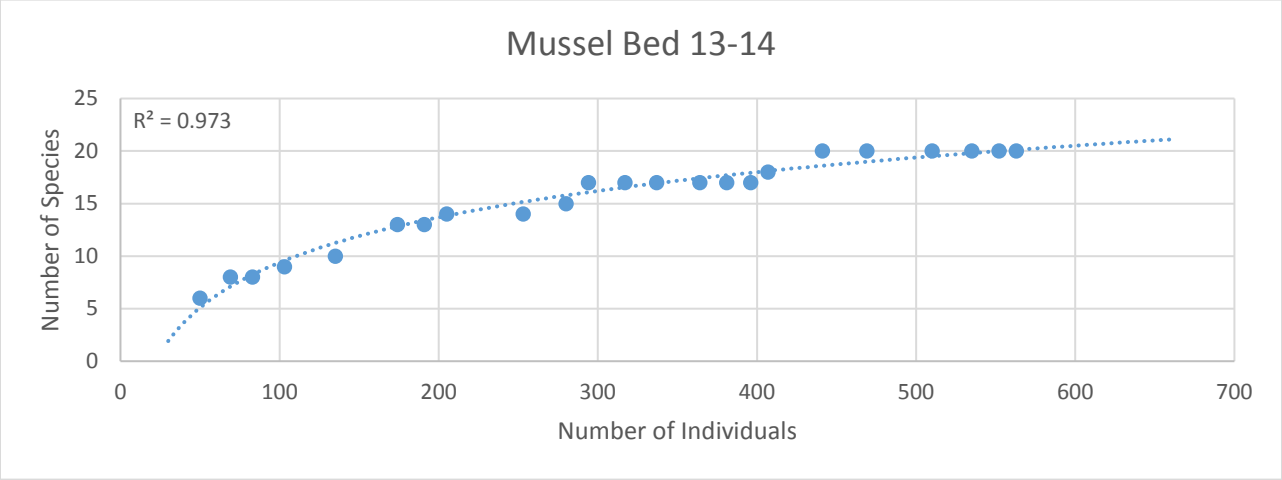
Tables 4A – 4N. Physical parameters, species composition, and population and community estimates of selected major mussel beds in the Little River, Arkansas – Oklahoma state line to U.S. Highway 71 near Millwood Lake, Arkansas, 2016.

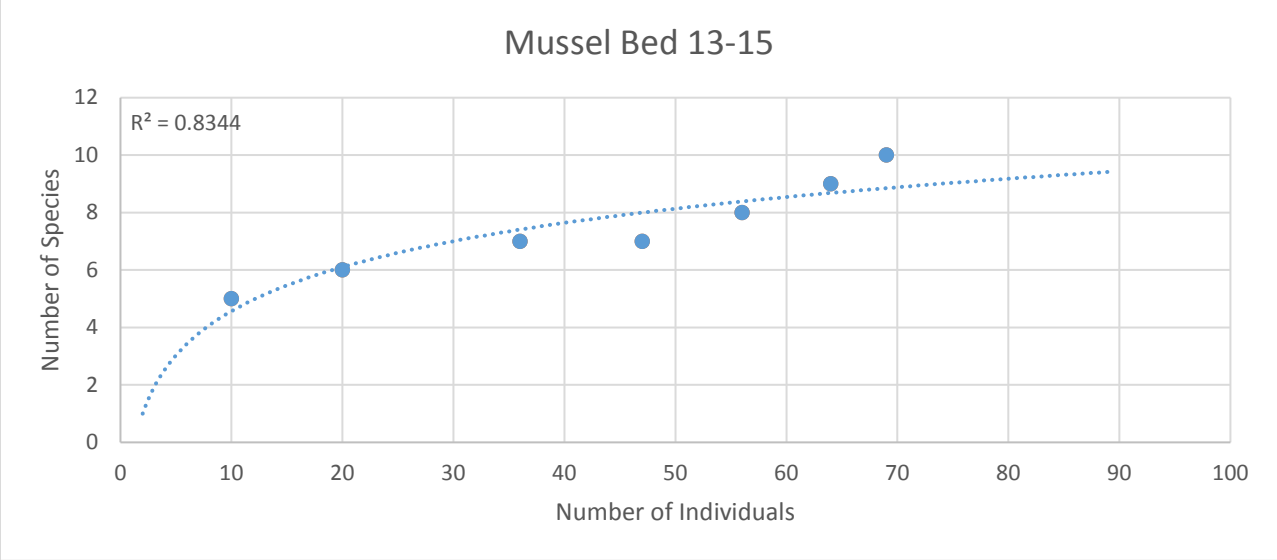
Appendix A14			
Location: 13-53		Date: 5/6 October 2016	
Estimated Mussel Bed Area (m^2):		1.) 2,130 2.) 1,600 3.) 8,400 Total: 12,130	
1 m^2 Quadrats Sampled:		40	
Min – Max Density (#/m^2):		12 – 70	
Mean Density #/m^2 (SD):		1.) 31.9 (21.8) 2.) 18.4 (3.6) 3.) 26.9 (8.0) Total: 26.7 (11.5)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	264	24.7	80,058 ± 11,965
Amblema plicata	6	0.6	1,820 ± 1,053
Arkansia wheeleri	1	0.1	303 ± 471
Ellipsaria lineolata	35	3.3	10,614 ± 3,649
Fusconaia flava	12	1.1	3,639 ± 1,825
Lampsilis cardium	4	0.4	1213 ± 999
Megalonaias nervosa	4	0.4	1213 ± 999
Obliquaria reflexa	155	14.5	47,004 ± 12,859
Obovaria arkansasensis	5	0.5	1,516 ± 1,053
Plectomerus dombeyanus	2	0.2	307 ±745
Pleurobema ridellii	7	0.7	2,123 ± 1,490
Pleurobema rubrum	8	0.7	2426 ± 1,490
Pleurobema sintoxia	8	0.7	2,426 ± 1,825
Potamilus purpuratus	1	0.1	303 ± 471
Ptychobranchus occidentalis	3	0.3	910 ± 881
Qudrula cylindrica cylindrica	4	0.4	1,213 ± 1,053
Quadrula pustulosa	465	43.5	141,011 ± 19,253
Quadrula verrucosa	7	0.7	2,123 ± 1,053
Truncilla truncata	77	7.2	23,350 ± 4,080
Totals	1,068	100.1	323,871 ± 38,433

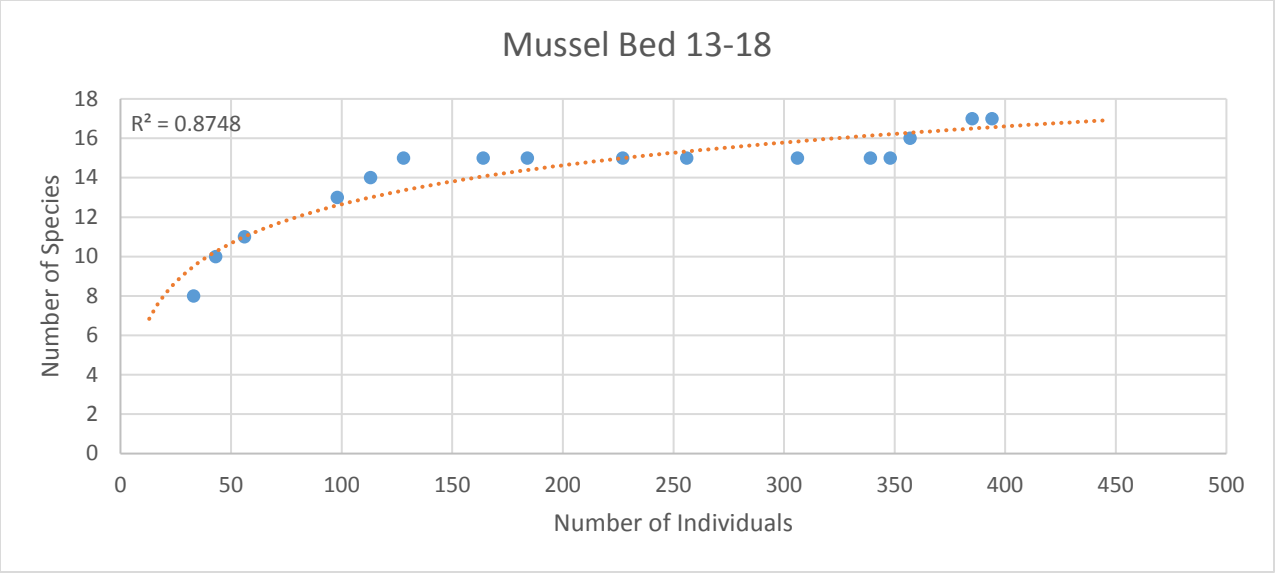
Appendix B. Results of quantitative data for Mussel Bed 13-27 in the Little River, Arkansas – Oklahoma state line to Millwood Lake, 2008.

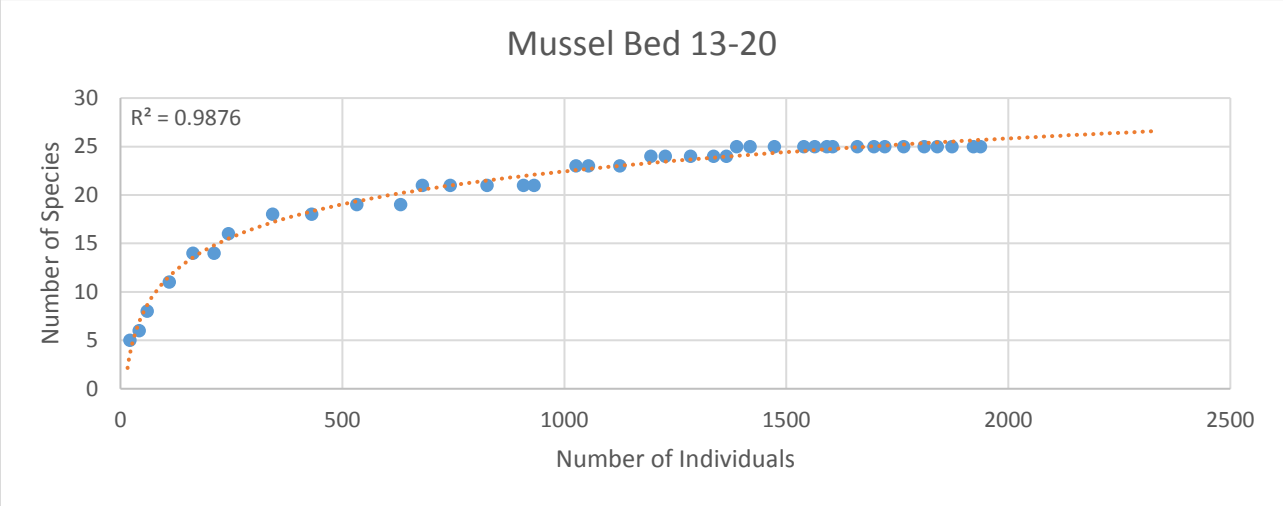
Location: 13-27		Date: 31 July 2008	
Estimated Mussel Bed Area (m^2):		15,525	
1 m^2 Quadrats Sampled:		37	
Min – Max Density (#/m^2):		5 – 78	
Mean Density #/m^2 (SD):		28.8 (16.5)	
Species	**Number Collected**	**Percent of Total**	**Population Estimate**
Actinonaias ligamentina	159	14.9	65,037 ± 21,855
Amblema plicata	10	0.9	4,196 ± 2,895
Arkansia wheeleri	1	0.1	419 ± 730
Ellipsaria lineolata	11	1.0	4,616 ± 2,309
Fusconaia flava	30	2.8	12,588 ± 4,296
Lampsilis cardium	10	0.9	4,196 ± 2,895
Leptodea fragilis	3	0.3	1,259 ± 1,614
Megalonaias nervosa	9	0.8	3,776 ± 2,176
Obliquaria reflexa	114	10.7	47,833 ± 11.909
Obovaria arkansasensis	1	0.1	419 ± 730
Plectomerus dombeyanus	4	0.4	1,678 ± 1,404
Pleurobema ridellii	11	1.0	4,616 ± 2,309
Pleurobema sintoxia	2	0.2	839 ± 1,460
Ptychobranchus occidentalis	3	0.3	1,259 ± 1,614
Quadrula apiculata	3	0.3	1,259 ± 1,614
Qudrula cylindrica cylindrica	5	0.5	2,098 ± 1,539
Quadrula pustulosa	484	45.4	203,084 ± 34,868
Reginaia ebena	1	0.1	419 ± 730
Tritogonia verrucosa	31	2.9	13,007 ± 5,390
Truncilla truncata	169	15.8	70,911 ± 13,791
Totals	1,061	100.0	447,707 ± 73,066

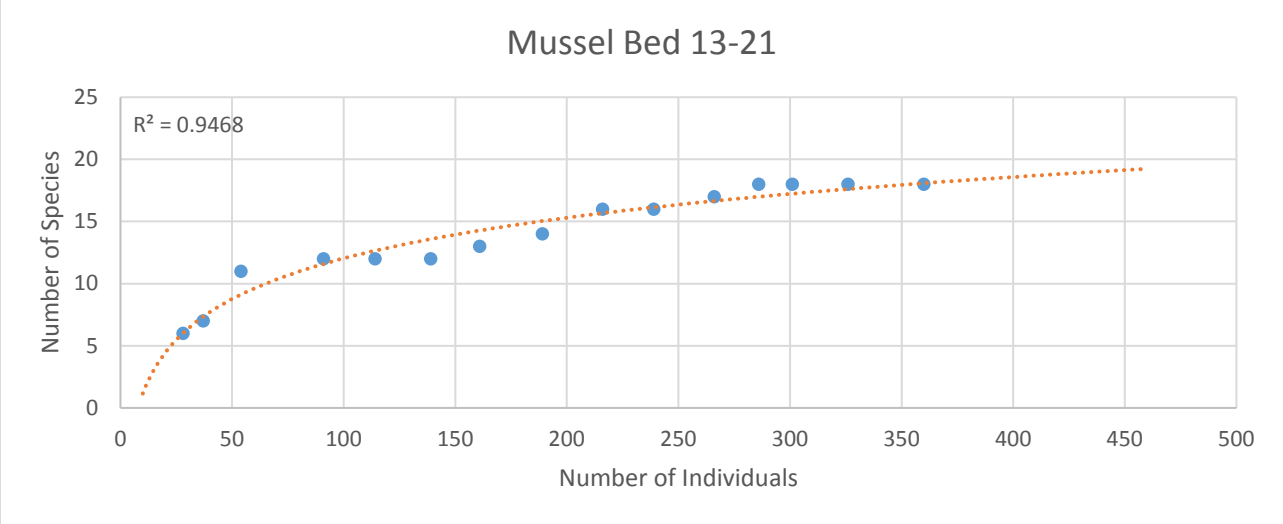
Appendix C. Regression of cumulative number of species versus the log of cumulative individuals for selected mussel beds in the Little River to predict effort required to obtain additional species.

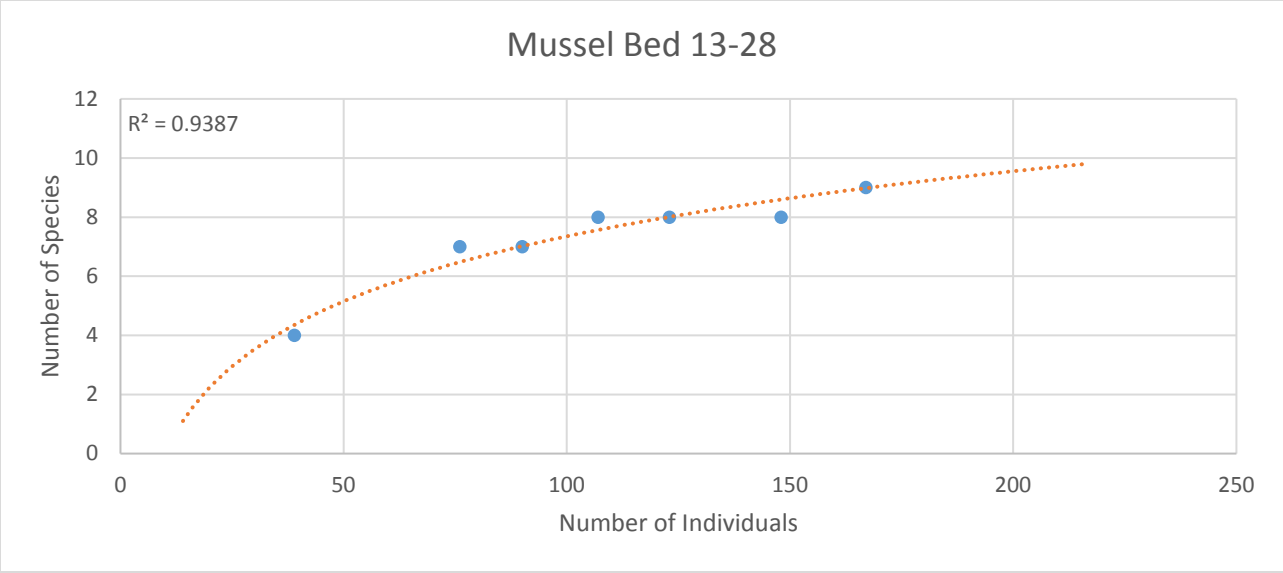


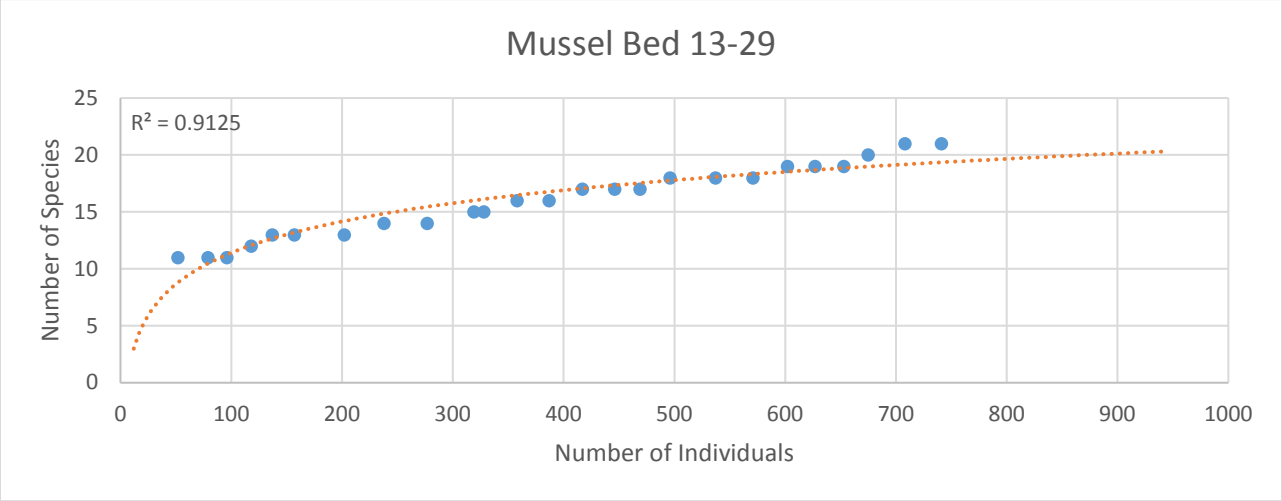


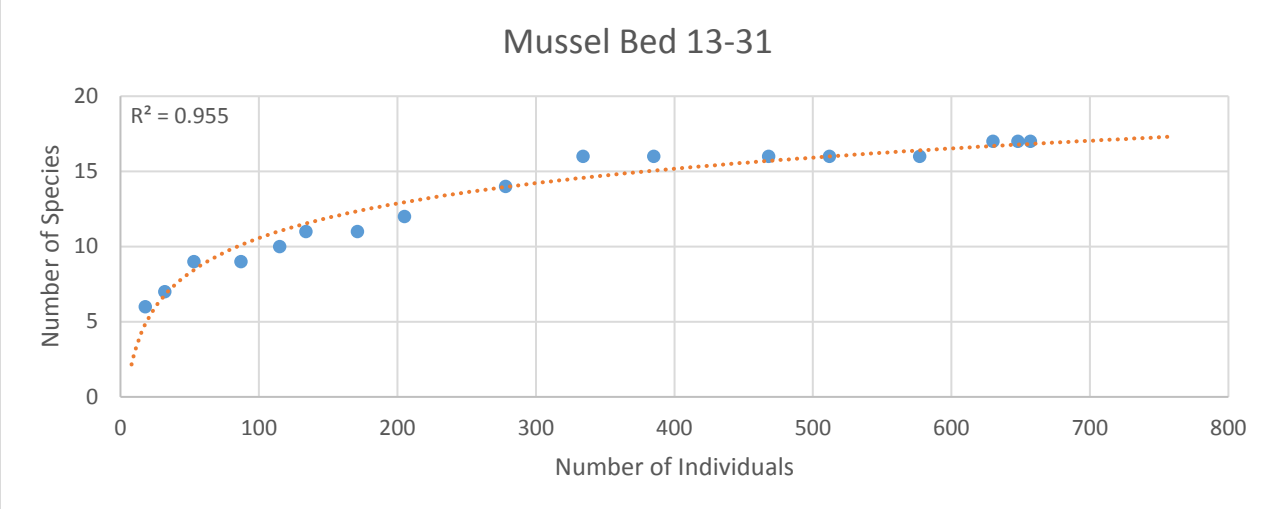

Mussel Bed 13-18
R² = 0.8748
Number of Species
Number of Individuals
0
2
4
6
8
10
12
14
16
18
0
50
100
150
200
250
300
350
400
450
500

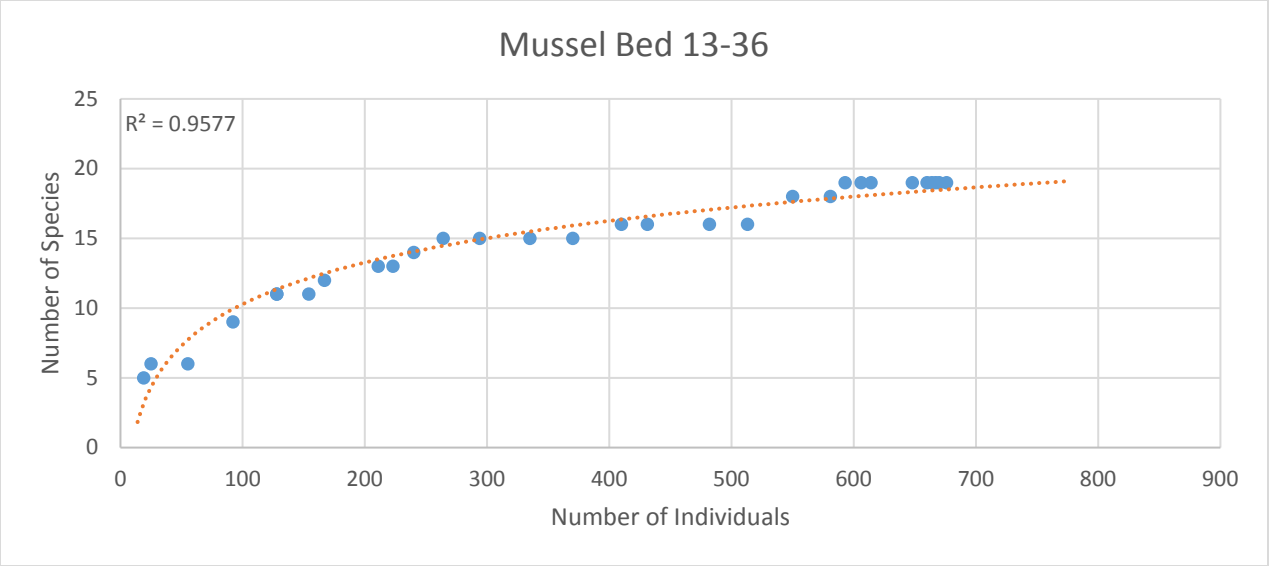

Mussel Bed 13-20
R² = 0.9876
Number of Species
Number of Individuals
0
5
10
15
20
25
30
0
500
1000
1500
2000
2500

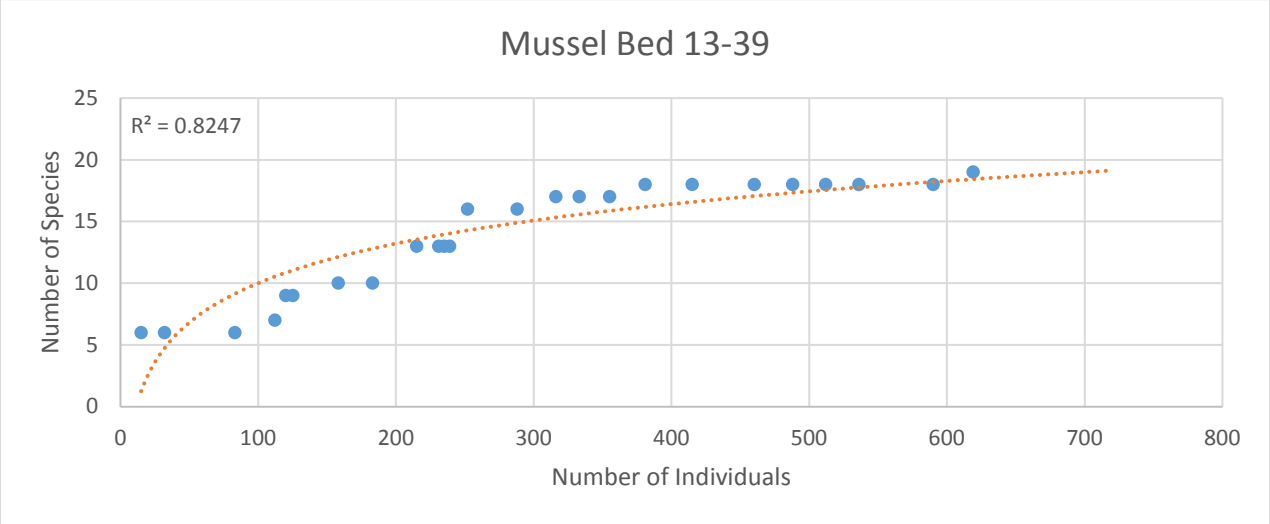

Mussel Bed 13-21
R² = 0.9468
Number of Species
Number of Individuals
0
5
10
15
20
25
0
50
100
150
200
250
300
350
400
450
500

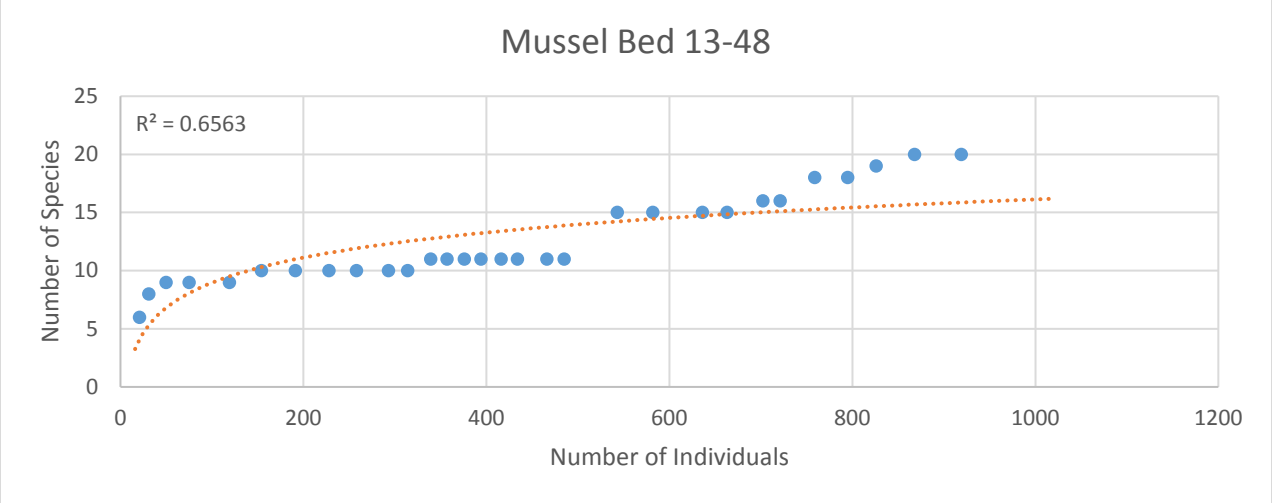

Mussel Bed 13-28
R² = 0.9387
Number of Species
Number of Individuals
0
2
4
6
8
10
12
0
50
100
150
200
250

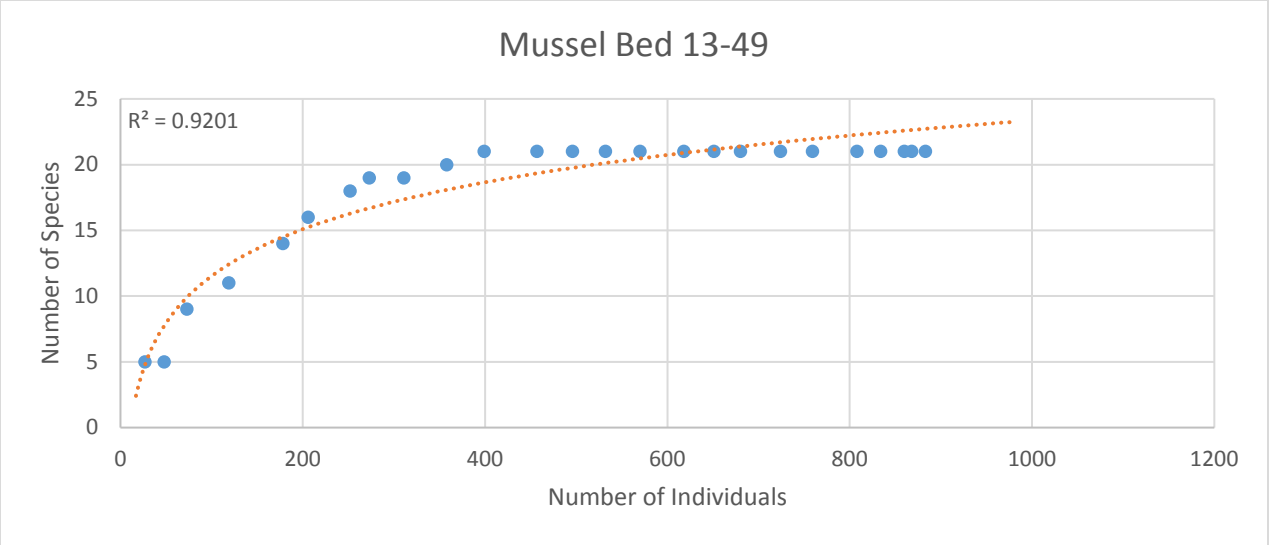

Mussel Bed 13-29
R² = 0.9125
Number of Species
Number of Individuals
0
5
10
15
20
25
0
100
200
300
400
500
600
700
800
900
1000

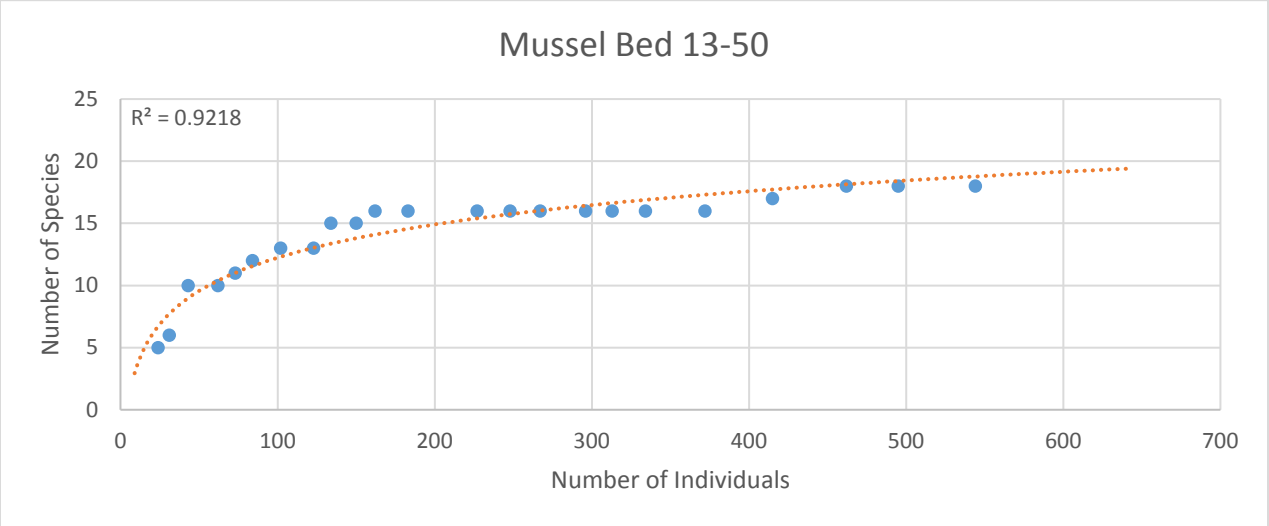

Mussel Bed 13-31
R² = 0.955
Number of Species
Number of Individuals
0
5
10
15
20
0
100
200
300
400
500
600
700
800

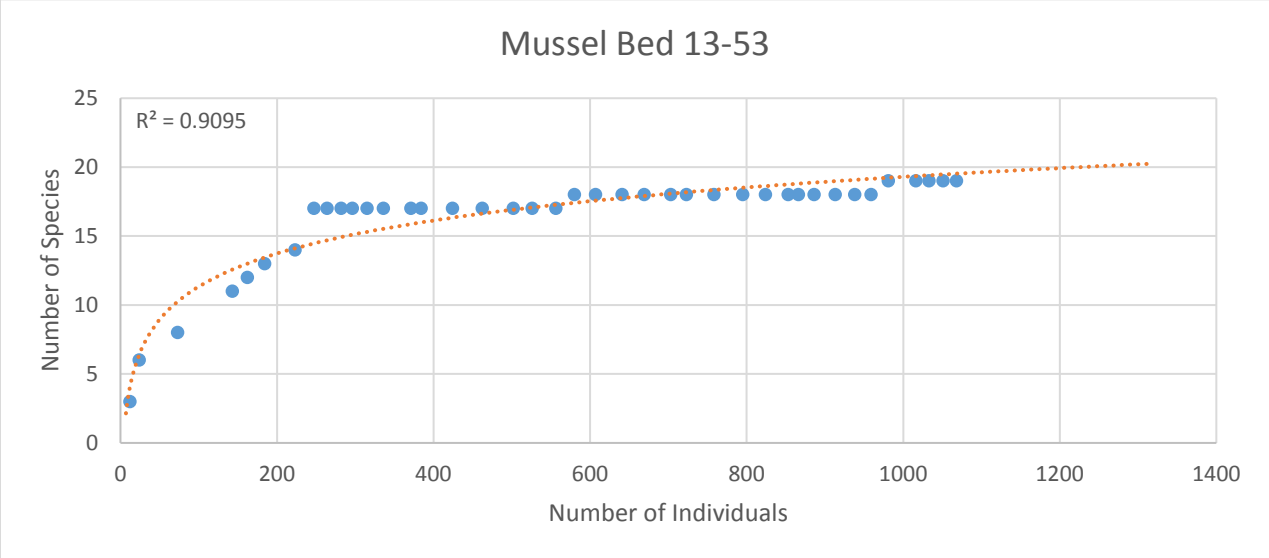

Mussel Bed 13-36
R² = 0.9577
Number of Species
Number of Individuals
0
5
10
15
20
25
0
100
200
300
400
500
600
700
800
900


Mussel Bed 13-39
R² = 0.8247
Number of Species
Number of Individuals
0
5
10
15
20
25
0
100
200
300
400
500
600
700
800


Mussel Bed 13-48
R² = 0.6563
Number of Species
Number of Individuals
0
5
10
15
20
25
0
200
400
600
800
1000
1200


Mussel Bed 13-49
R² = 0.9201
Number of Species
Number of Individuals
0
5
10
15
20
25
0
200
400
600
800
1000
1200


Mussel Bed 13-50
R² = 0.9218
Number of Species
Number of Individuals
0
5
10
15
20
25
0
100
200
300
400
500
600
700


Mussel Bed 13-53
R² = 0.9095
Number of Species
Number of Individuals
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400